$$\xrightarrow[+3\times \text{Row}1]{\text{Row}2=\text{Row}2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \\ 2 & -4 & 5 & 8 & -4 & 0 \end{bmatrix} \xrightarrow[-2\times \text{Row}1]{\text{Row}3=\text{Row}3}$$

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 5 & 10 & -10 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \end{bmatrix} \xrightarrow[\times 1/5]{\text{Row}2=\text{Row}2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \end{bmatrix}$$

$$\xrightarrow[\text{Row}2]{\text{Row}3=\text{Row}3-} \begin{bmatrix} \boxed{1} & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & \boxed{1} & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-2\times \text{Row}2]{\text{Row}1=\text{Row}1} \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

It's very easy to verify that the final matrix is in echelon form. Row 1 has its pivot in Column 1 and Row 2 has its pivot in Column 3. Column 2, 4 and 5 has no pivot, i.e. x_2, x_4 and x_5 are free variables.

By Theorem 2 from Chapter 1, the system of equations has infinite solutions. By Defn, the columns of A are linearly dependent.

In parametric vector form, the solution is:

$$x_1 - 2x_2 - x_4 + 3x_5 = 0$$
$$x_3 + 2x_4 - 2x_5 = 0$$
$$0 = 0$$

The general solution is $x_1 = 2x_2 + x_4 - 3x_5$, $x_3 = -2x_4 + 2x_5$, with x_2, x_4 and x_5 free.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{u} + x_4 \underbrace{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{v} + x_5 \underbrace{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}_{w}$$

$$= x_2 u + x_4 v + x_5 w \quad (1)$$

Equation (1) shows that Nul A coincides with the set of all linear combinations of u, v and w. That is, $\{u, v, w\}$ generates Nul A. In fact, this construction of u, v and w automatically makes them linearly independent, because equation (1) shows that $0 = x_2 u + x_4 v + x_5 w$ only if the weights x_2, x_4 and x_5 are all zero. So $\{u, v, w\}$ is a **basis** for Nul A. □